

Effects of very-low-carbohydrate (horsemeat- or beef-based) diets and restricted feeding on weight gain, feed and energy efficiency, as well as ser...

BACKGROUND/AIMS: The beneficial or harmful effect of the low-carbohydrate (low-carb), high-protein, high-fat diet (Atkins diet) has not been clearly demonstrated. We determined the effect of a low-carb diet and restricted feeding (70% ad libitum intake) on serum levels of cholesterol, triacylglycerol, glucose, ketone bodies and insulin in rats. **METHODS:** In experiment 1, each of 4 groups with 10 adult rats was assigned to a high-carb diet (AIN-93G) + ad libitum intake or restricted feeding, or a low-carb diet (53% horsemeat) + ad libitum intake or restricted feeding (2 x 2 factorial). In experiment 2, each of 3 groups with 10 adult rats was assigned to a control (AIN-93G) or low-carb diets (53% beef or horsemeat). **RESULTS:** Restricted feeding and the low-carb diet reduced ($p < 0.01$) serum triacylglycerol compared with ad libitum intake and the AIN-93G diet, respectively (experiment 1). The dietary effect on serum total cholesterol, high-density or low-density lipid cholesterol appeared to be inconsistent, but restricted feeding increased the low-density lipoprotein cholesterol level. The serum ketone body level was increased by the low-carb diet compared with AIN-93G (experiment 2). **CONCLUSION:** Restricted feeding and a low-carb diet are beneficial for alleviating cardiovascular disease risk factors, and their effects are additive, restricted feeding being more pronounced. Copyright 2009 S. Karger AG, Basel.